RAJALAKSHMI ENGINEERING COLLEGE
Approved by AICTE | Affiliated to Anna University | Accredited by NAAC

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence
V Semester B. Tech AIDS

AD23532 – Principles of Data Science

Academic Year: 2026-2027 **Batch: 2024-2028**

Ex1: Data Preprocessing and Cleaning

Aim: To clean and preprocess the Telco Customer Churn dataset by handling duplicates and missing values, encoding and scaling the features, and visualising the data.

Problem Statement:
A telecommunications company is experiencing customer churn (customers leaving the service). Before building a machine learning model to predict churn, the data science team must clean and preprocess the customer data to ensure high-quality input for analysis.

You are a Data Scientist working for the company. Your task is to preprocess the customer dataset by handling missing values, removing inconsistencies, transforming categorical features, scaling numerical attributes, and visualizing important relationships.

Learning Objective:
The objective of this experiment is to enable students to understand and perform the essential steps involved in data preprocessing. This experiment will provide a strong foundation for preparing high-quality data for subsequent machine learning and predictive analytics tasks.

Tasks to Perform:

1. Import the required Python libraries.
2. Load the Telco Customer Churn Dataset into a Pandas DataFrame. And display First 10 records, Last 10 records, Shape of the dataset, Column names.
3. Explore the Dataset and display Dataset information, Summary statistics, Number of missing values, Number of duplicate records
4. Clean the data
    a. Remove duplicate customer records.
    b. Replace missing values
        i. Numerical columns → Median
        ii. Categorical columns → Mode
    c. Verify that no missing values remain.
5. Encode categorical columns: Gender, Partner, Dependents, Contract, PaymentMethod, InternetService, Churn. Use appropriate encoding techniques.
6. Standardise the numerical features: Tenure, MonthlyCharges, TotalCharges, using StandardScaler.
7. Plot Distribution of MonthlyCharges, Distribution of Tenure.
8. Create a Count Plot showing Contract vs Churn.
9. Draw a Box Plot of MonthlyCharges grouped by Churn.
10. Display the Correlation Heatmap of all numerical features.

Solution:

```
# 1. Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```



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```
from sklearn.preprocessing import LabelEncoder, StandardScaler

# 2. Load dataset
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
print(df.head(10))
print(df.tail(10))
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())

# 3. Explore dataset
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df.info()
print(df.describe())
print("Missing values:\n", df.isnull().sum())
print("Duplicate records:", df.duplicated().sum())

# 4. Clean data
df = df.drop_duplicates()
num_cols = df.select_dtypes(include=np.number).columns
cat_cols = df.select_dtypes(exclude=np.number).columns.drop("customerID")
for c in num_cols:
    df[c] = df[c].fillna(df[c].median())      # numerical -> median
for c in cat_cols:
    df[c] = df[c].fillna(df[c].mode()[0])     # categorical -> mode
print("Missing values left:", df.isnull().sum().sum())
df_clean = df.copy()                          # keep readable labels for plots

# 5. Encoding
for c in ["gender", "Partner", "Dependents", "Churn"]:      # binary -> Label
    df[c] = LabelEncoder().fit_transform(df[c])
df = pd.get_dummies(df, columns=["Contract", "PaymentMethod",
                                 "InternetService"], dtype=int)  # One-Hot

# 6. Standardise numerical features
num = ["tenure", "MonthlyCharges", "TotalCharges"]
df[num] = StandardScaler().fit_transform(df[num])
print(df[num].describe().round(2))

# 7-8. Distributions
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
sns.histplot(df_clean["MonthlyCharges"], kde=True, ax=ax[0])
sns.histplot(df_clean["tenure"], kde=True, ax=ax[1])
ax[0].set_title("Distribution of MonthlyCharges")
ax[1].set_title("Distribution of Tenure")
plt.show()

# 9. Count plot: Contract vs Churn
sns.countplot(data=df_clean, x="Contract", hue="Churn")
plt.title("Contract vs Churn")
plt.show()

# 10. Box plot: MonthlyCharges by Churn
sns.boxplot(data=df_clean, x="Churn", y="MonthlyCharges")
plt.title("MonthlyCharges by Churn")
plt.show()

# 11. Correlation heatmap
corr = df[["SeniorCitizen", "tenure", "MonthlyCharges",
           "TotalCharges", "Churn"]].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```

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Result:

The Telco dataset (7,043 rows, 21 columns) was cleaned and preprocessed. There were no duplicates and the 11 missing TotalCharges values were filled with the median. The categorical columns were encoded, the numerical columns were standardised and all the plots were generated.